Falcon Ledger Lending — Implementation Report

Date: 2026-07-14
Network: Falcon Ledger Testnet (network ID `1001`, RPC `http://46.224.0.140:6005`)
Status: End-to-end supply, permissionless borrow, repay, and liquidation verified on-ledger (coordinator E2E scripts + portal `/lend`). HF monitor daemon deployed on coordinator.

> **Borrowers do not need a broker.** With `LendingPermissionless` enabled on testnet (live since ledger ~126464), borrow is **collateral-only**: post FALCON at ≥150% health factor, sign `LoanSet` yourself — no broker co-sign, no broker first-loss cover, no server secret. Legacy broker co-sign below is **historical only** (pre-permissionless).

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1. Executive summary

Falcon Ledger lending is built on upstream **XRPL XLS-66** primitives (`SingleAssetVault`, `LendingProtocol`, `MPTokensV1`) plus qXRP extensions:

- **`ClaimLPReward`** — Proof-of-Participation LP epoch emissions (FALCON)
- **`LendingCollateral`** — FALCON collateral locked in `LoanSet`
- **`LendingPermissionless`** — collateral-only borrow without broker `CounterpartySignature`; permissionless liquidation

The **protocol implementation** lives in the `qXRP` repository; the **user-facing lending loop** is wired in the `qXRP-faucet-wallet` portal at `/lend`.

Capability	Portal	On-chain
Supply F-USDC to vault	□ `VaultDeposit`	□
Borrow F-USDC (permissionless — **current**)	□ `LoanSet` + FALCON collateral	□ `LendingPermissionless` live on testnet
Borrow F-USDC (legacy broker — **retired**)	Optional co-sign path in code	Pre-July 2026 only; not required
Repay loan	□ `LoanPay` (Pay full amount)	□
Withdraw supply	□ `VaultWithdraw`	□
Claim LP epoch rewards	□ `ClaimLPReward`	□ FALCON (PoPL)
FALCON collateral in `LoanSet`	□	□ `LendingCollateral`
Health factor display (AMM price)	□ borrow preview + Positions + risk monitor	UI + daemon
On-chain liquidation / impairment	□ `LoanManage` + HF monitor daemon	□ anyone can default on HF breach or late payment
Borrow / repay / claim / withdraw preflight	□ `/api/lend/*-preflight`	simulate before sign
Multi-loan Positions UI	□ loan selector	filters paid/closed loans

Economics

- **LP interest:** `LoanPay` returns principal + interest in **F-USDC** to the vault. LP share value rises — there is no separate interest claim transaction.
- **LP emissions:** `ClaimLPReward` distributes **FALCON** from the epoch PoPL participation split.
- **Liquidation:** on default, the **liquidator receives FALCON collateral**. The vault books collateral value at AMM price (`collateralVaultValue`); any residual shortfall reduces vault F-USDC accounting. LPs are not paid out in FALCON on default.

Liquidity in the lend pool is **real F-USDC** (QUC IOU from issuer `rsJoDhjVV78jr6huHxKjtT8uG8RGeGmd1N`), not bootstrap-minted supply.

Verified permissionless E2E (2026-07-14): borrow **5 F-USDC** + repay **5.000069** (`tesSUCCESS`); HF-breach liquidation with third-party `LoanManage` default (`tesSUCCESS`). See §7.

Historical (legacy broker path, 2026-07-10): wallet `rwcYXAAXe7unkEwPVFWMbyzXE2ajG3juqR` — borrow 10 F-USDC with broker co-sign; repay **10.000137** F-USDC.

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2. Architecture

```
flowchart TB
  subgraph Portal["Falcon portal (qXRP-faucet-wallet)"]
    LendUI["/lend UI"]
    OverviewAPI["GET /api/lend/overview"]
    PreflightAPI["POST /api/lend/*-preflight"]
    CosignAPI["POST /api/lend/cosign (legacy)"]
    LoanManageAPI["POST /api/lend/loan-manage"]
    SubmitAPI["POST /api/wallet/submit"]
    SignClient["falcon-lend-tx-sign.ts (browser WASM)"]
  end

  subgraph Server["Vercel / Node server"]
    SignerProxy["Signer proxy (node1 :3001)"]
    BrokerSecret["TESTNET_LENDING_BROKER_SECRET (legacy)"]
    HFMonitor["lend-hf-monitor.py (coordinator)"]
  end

  subgraph Chain["Falcon Ledger (xrpld)"]
    Vault["SingleAssetVault"]
    Broker["LoanBroker"]
    Loan["Loan objects"]
    POP["ClaimLPReward (PoP)"]
    Perm["LendingPermissionless"]
  end

  LendUI --> SignClient
  LendUI --> OverviewAPI
  LendUI --> PreflightAPI
  LendUI --> SubmitAPI
  LendUI -.-> CosignAPI
  OverviewAPI --> Chain
  CosignAPI --> SignerProxy
  SignerProxy --> BrokerSecret
  HFMonitor --> LoanManageAPI
  LoanManageAPI --> SignerProxy
  SubmitAPI --> Chain
  SignClient --> SubmitAPI
  Vault --> Broker
  Broker --> Loan
  Vault --> POP
  Perm --> Loan
```

Repositories

Repo	Role
`qXRP`	xrpld fork: transactors, `LendingHelpers`, `LendingPermissionless`, HF monitor scripts, fleet amendment enablement
`qXRP-faucet-wallet`	Portal: `/lend` UI, overview aggregation, preflight APIs, client tx signing, optional broker co-sign

Amendments (enable order)

1. **MPTokensV1** — vault share tokens (MPT) for LP positions
2. **SingleAssetVault** — `VaultCreate`, `VaultDeposit`, `VaultWithdraw`
3. **LendingProtocol** — `LoanBrokerSet`, `LoanSet`, `LoanPay`, `LoanManage`
4. **LendingCollateral** — `Collateral` field on `LoanSet`; FALCON locked on-chain

5. **LendingPermissionless** — collateral-only borrow; permissionless `LoanManage` default

Fleet scripts: `qXRP/scripts/enable-lending-fleet.sh`, `qXRP/scripts/enable-lending-permissionless-fleet.sh`.

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3. On-chain objects (testnet)

Authoritative manifest: `qXRP-faucet-wallet/public/config/lending.json`

Item	Value
Network ID	`1001`
F-USDC currency	`QUC`
F-USDC issuer	`rsJoDhjVV78jr6huHxKjtT8uG8RGeGmd1N`
Vault ID	`0DB363B417A560EDD7EA8306188F5592F2388A054BF7F6AC1FB5A99A30BC99B2`
Loan broker ID	`0DF028DFE8928921B9474B5EB09531E1E7A3655441C53ECFECF41C82F374D334`
Broker owner (legacy)	`rJePmBhHoerhB4gJPAPEqvVBgQ7xbmY6bh` — not a borrow gate once permissionless is live
Interest	`500` tenth-bips = **5% APR**
Payment interval	`86400` s (1 day)
Payment total	`1` (single-installment test loans)
Grace period	`3600` s (1 hour)

Collateral constants (`LendingHelpers.h`)

Constant	Value	Meaning
`kPermissionlessMinCollateralBps`	15000	Min HF 1.5 at borrow (150% collateralization)
`kPermissionlessLiquidationHfBps`	11000	Liquidatable when HF < 1.1

Price source: FALCON/F-USDC AMM mid. Loans without broker co-sign carry `lsfLoanPermissionless`.

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4. Protocol rules

4.1 Vault (supply side)

- **VaultDeposit** — LP sends F-USDC; receives vault share MPT.
- **VaultWithdraw** — LP burns share MPT; receives F-USDC (FCFS).
- **ClaimLPReward** — LP claims FALCON epoch emission by vault share balance.

4.2 Permissionless borrow (`LendingPermissionless` + `LendingCollateral`)

- **LoanSet** — borrower signs only; includes `Collateral` (FALCON drops); no `CounterpartySignature` .
- `Lending::checkPermissionlessCollateral` enforces HF ≥ 1.5 at AMM price.
- Broker cover check is skipped; loan flagged `lsfLoanPermissionless` .

4.3 Legacy broker borrow (retired — historical reference only)

> Not used on testnet after `LendingPermissionless` fleet enable. Documented for protocol completeness only.

- **LoanSet** — previously required broker `CounterpartySignature` and `CoverAvailable ≥ 1% × debt` .
- **LoanBrokerCoverDeposit** — operator posted F-USDC first-loss cover.

4.4 Loan lifecycle

- **LoanPay** — installment must be ≥ periodic payment (principal + interest/fees). Principal-only → `tecINSUFFICIENT_PAYMENT` .
- **LoanManage (permissionless)** — any account may:
  - `impair` when HF < 1.1
  - `default` when HF < 1.1 **or** payment past grace period
- **Default settlement** (`defaultPermissionlessLoan`):
  1. Transfer FALCON collateral to liquidator
  2. Credit vault `AssetsAvailable` by collateral value at AMM price (up to debt owed)
  3. Reduce vault `AssetsTotal` by residual F-USDC shortfall

4.5 Rate encoding

1 tenth-bip = 0.0001%; 100,000 tenth-bips = 100%.

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5. Portal implementation

5.1 UI (`/lend`)

Tab	Panel	Transaction
Overview	Pool stats, APY, risk monitor	Read-only
Supply	`LendSupplyPanel`	`VaultDeposit`
Borrow	`LendBorrowPanel`	`LoanSet`
Positions	`LendPositionsPanel`	`VaultWithdraw` , `LoanPay` , `ClaimLPReward`

When `protocol.lendingPermissionless && protocol.lendingCollateral` , borrow skips co-sign and shows permissionless copy.

5.2 API routes

Route	Purpose
`GET /api/lend/overview`	Vault, broker, epoch/PoP, AMM price, loans, LP positions, amendment flags
`POST /api/lend/borrow-preflight`	Collateral HF, vault liquidity, cosign requirement
`POST /api/lend/repay-preflight`	Installment + wallet balance
`POST /api/lend/claim-preflight`	Claim eligibility
`POST /api/lend/withdraw-preflight`	Share balance + vault utilization
`POST /api/lend/supply-preflight`	F-USDC balance + trust line
`GET /api/lend/risk-monitor`	Fleet-wide HF scan
`POST /api/lend/loan-manage`	HF daemon / ops: `impair` / `unimpair` / `default`
`POST /api/lend/cosign`	Legacy testnet broker co-sign
`POST /api/wallet/submit`	Submit signed `tx_blob`

5.3 Client transaction signing (`falcon-lend-tx-sign.ts`)

Function	Transaction
`signVaultDepositTx`	`VaultDeposit`
`signVaultWithdrawTx`	`VaultWithdraw`
`signClaimLPRewardTx`	`ClaimLPReward`
`signLoanSetBorrowerTx`	`LoanSet`
`signLoanPayTx`	`LoanPay`

5.4 Error UX (`lend-borrow-errors.ts`)

- `borrowBlockedReason` — skips broker cover when permissionless; checks cosign only on legacy path
- `repayBlockedReason` / `fullRepayAmount` — installment-aware repay
- `collateralBlockedReason` (`lend-collateral.ts`) — 150% min collateral at AMM price

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6. End-to-end flows

6.1 Supply

1. User holds F-USDC + trust line
2. Lend → Supply → passkey → `VaultDeposit`
3. Vault share MPT received; `AssetsAvailable` increases

6.2 Borrow (permissionless — target path)

1. Portal checks collateral HF ≥ 1.5, vault liquidity (`borrow-preflight`)
2. Borrower signs `LoanSet` with `Collateral` — no co-sign
3. Submit → F-USDC to borrower; FALCON locked on loan

6.3 Borrow (legacy broker — retired)

Historical path only. Permissionless borrowers **skip** broker cover and co-sign entirely.

6.4 Repay

1. UI auto-fills exact `PeriodicPayment` (e.g. `10.000137`)
2. `LoanPay` → F-USDC to vault (LP yield via share value)

6.5 Claim LP rewards

`ClaimLPReward` → FALCON from epoch emission.

6.6 Liquidation

`lend-hf-monitor.py` on coordinator scans loans → `POST /api/lend/loan-manage` with `default` or `impair` . Any account can also submit `LoanManage` on permissionless loans when eligible.

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7. End-to-end test program (coordinator)

All permissionless lending E2E tests run **on-ledger** against live testnet RPC (`http://46.224.0.140:6005`) from the **coordinator** (`root@46.224.0.140`) using admin RPC signing. No portal passkey or broker co-sign is required for these scripts.

7.1 Fleet and amendment status

Item	Value
Docker image	`qxrp/xrpld:lending-permissionless` (commits `ce6c03319` feature, `90d14366b` E2E fixes)
Fleet	7/7 validators upgraded via `bin/install/rolling-upgrade-fleet.sh`
`LendingPermissionless`	Enabled at flag ledger ~**126464** (5/5 votes + 15-min hold)
Coordinator container	`qxrp-full`
Validated ledger (report date)	~**130168**

7.2 E2E scripts (`qXRP/scripts/`)

Script	Purpose	Exit criteria
`lend-e2e-permissionless.py`	Supply (optional) → permissionless borrow → repay	All steps `tesSUCCESS`
`lend-e2e-liquidation.py`	Borrow → AMM price shock → `LoanManage` default	Liquidator receives FALCON collateral
`lend-hf-monitor.py`	HF + payment-default enforcement daemon	`LoanManage` when HF < 1.1 or late payment
`deploy-lend-hf-monitor.sh`	Install systemd unit `qxrp-lend-hf-monitor.service`	Service active on coordinator

Run (coordinator):

```
python3 /root/qXRP/scripts/lend-e2e-permissionless.py
python3 /root/qXRP/scripts/lend-e2e-liquidation.py
python3 /var/lib/qxrp-lending/lend-hf-monitor.py --once --dry-run
```

7.3 Test harness design

Each E2E script:

1. Loads state from `/var/lib/qxrp-stables/stables_state.json` , `lending_state.json` , `/root/qxrp-bootstrap/faucet.json`
2. Proposes fresh Falcon wallets via `wallet_propose` (admin RPC)
3. Funds wallets from faucet (`rwzhiWW4GYK2sQVR5Lw4iDpYLANB5krJXY` , ~390k FALCON)
4. Signs and submits via `docker exec qxrp-full curl` → admin `:5005` / public `:6005`
5. Waits for validated ledger confirmation per tx

Wallet roles (liquidation script):

Role	Source	Purpose
Borrower	`wallet_propose`	`LoanSet` permissionless borrow
Liquidator	`wallet_propose`	Third-party `LoanManage` default (proves no broker needed)
Price dumper	Faucet account	AMM swap: sell FALCON → F-USDC (`Payment` + `tfPartialPayment`)

Validator fleet accounts are **not** used for E2E — bonded FALCON and key isolation make the faucet the correct whale account on the coordinator.

7.4 Test A — Permissionless borrow + repay □

Script: `lend-e2e-permissionless.py`

Flow:

```
fund lender + borrower (2000 FALCON each)
→ trust lines (QUC / F-USDC)
→ issuer mint F-USDC (30 lender / 15 borrower)
→ VaultDeposit 20 F-USDC OR supply_skip if vault AssetsAvailable ≥ 10
→ LoanSet: 5 F-USDC principal, FALCON collateral @ 1.5 HF (+5% buffer)
→ LoanPay: full installment rounded UP to 6 dp
```

Loan terms (script):

Field	Value
`PrincipalRequested`	`5` F-USDC
`InterestRate`	`500` tenth-bips (5% APR)
`PaymentInterval`	`86400` s
`PaymentTotal`	`1`
`GracePeriod`	`3600` s
`Flags`	`tfLoanOverpayment` (`0x00010000`)
Collateral	`ceil(5 × 1.5 / AMM_price × 1.05)` FALCON

Representative PASS run (2026-07-14):

Wallet	Address
Lender	`rUW5jpzLpEbfZ9GwpUk4Gs9iqoon4zTtY8`
Borrower	`rPxzyo4FdL7Pt7LekxpvTLTK2bLHZQBum8`

Step	Tx hash	Result
Borrow 5 F-USDC (875 FALCON collateral)	`78E3A2528B1C40FD09082D6249B65FC6EE1ECE9FE6F8B106E3E20F3FF81AC172`	tesSUCCESS
Repay **5.000069** F-USDC	`93B058510F90402CD34F12BF83C83BBDF684C6D2F8E06749AF3310892097E719`	tesSUCCESS

Repay lesson: `PeriodicPayment` on-chain is `5.000068493150794935` . Paying principal only (`5`) or truncated interest → `tecINSUFFICIENT_PAYMENT` . Portal and script use **ceil to 6 decimal places** (`5.000069`) — matches “Pay full amount” in UI.

Supply skip: When vault `AssetsAvailable` ≥ principal + buffer, script skips `VaultDeposit` to avoid intermittent `tecINVARIANT_FAILED` on repeated small deposits into a large vault (~255 F-USDC available at time of testing).

7.5 Test B — Permissionless liquidation (HF breach) ✅

Script: `lend-e2e-liquidation.py`

Flow:

```
fund borrower + liquidator
→ borrower trust line
→ LoanSet @ minimum 1.5 HF (no 5% buffer)
→ faucet trust line (receive F-USDC from AMM dumps)
→ faucet AMM dumps (sell FALCON for F-USDC until HF < 1.1)
→ liquidator LoanManage tfLoanDefault
→ assert lsfLoanDefault + liquidator FALCON balance += collateral
```

Health factor math (matches on-chain `loanHealthFactorBps`):

```
HF = (collateral_FALCON × AMM_FUSDC_per_FALCON) / debt_FUSDC
Liquidatable when HF < 1.1  (11,000 bps)
```

AMM dump mechanism: Self-`Payment` from faucet to faucet:

- `SendMax`: FALCON drops to sell
- `Amount`: max F-USDC to receive (must be large — **not** `1` USDC or price barely moves)
- `DeliverMin`: slippage floor
- `Flags`: `tfPartialPayment` (`0x00020000`)

Representative PASS run (2026-07-14, latest):

Wallet	Address
Borrower	`r3XK65UbcEhsif3UjWbqNdKeD28TBeSc62`
Liquidator	`rJM5vq5umHp82iztWHsZySAoUSMuD5VbgT`
Price dumper	`rwzhiWW4GYK2sQVR5Lw4iDpYLANB5krJXY` (faucet)

Step	HF / price	Tx hash	Result
Borrow 5 F-USDC, 1928 FALCON	HF **1.501** @ 0.003892	`0834FB47C0FB9CDE0D32E57FCD7666D54AD31A972BDB07FDF97C3601A63ED645`	tesSUCCESS
AMM dump 4000 FALCON	HF **1.231** @ 0.003194	`43E459BC0FB5C36B5D0CFAE7053E29DBDD0E40A72A4B7F5C8603B17D075A1984`	tesSUCCESS
AMM dump 8000 FALCON	HF **0.856** @ 0.002219	`989B04758E4BC7AFE598B73496DBCE48ABDA99CF002BFFF854E8898514734DBE`	tesSUCCESS
Liquidator default	—	`BA1C8431CEBDE5812DC9315EDE6B56FAAB4FCF0A592C0911303A3949DFA6588D`	tesSUCCESS

Post-liquidation: Liquidator FALCON **2000 → 3928** (+1928 collateral). Loan `AE6D230E…` flagged `lsfLoanDefault`.

On-chain settlement (`defaultPermissionlessLoan` in `LoanManage.cpp`):

1. FALCON collateral transferred from broker pseudo-account to **liquidator** (`account_` on `LoanManage`)
2. Vault `AssetsAvailable` credited by collateral value at AMM price (capped at debt owed)
3. Vault `AssetsTotal` reduced by any F-USDC shortfall
4. Loan debt fields zeroed; `lsfLoanDefault` set

7.6 HF monitor daemon □

Service: `qxrp-lend-hf-monitor.service` on coordinator
Binary: `/var/lib/qxrp-lending/lend-hf-monitor.py`
Interval: 60 s loop
State: `/var/lib/qxrp-lending/hf_monitor_state.json`

Actions (permissionless loans):

Condition	`LoanManage` flag
HF < 1.1	`tfLoanImpair`
HF < 1.1 or payment past grace	`tfLoanDefault`
Impaired + HF ≥ 1.1	`tfLoanUnimpair`

Broker secret loaded from `stables_state.json` `liquidity_provider.falcon_secret` or `TESTNET_LENDING_BROKER_SECRET` . For permissionless default, **any account** may submit — daemon uses broker owner for legacy loans; E2E uses a random liquidator wallet.

7.7 Portal session (2026-07-10, legacy broker path)

Wallet: `rwcYXAAXe7unkEwPVFWMbyzXE2ajG3juqR`

Step	Result	Notes
Vault deposit	□	100 F-USDC
AMM create	□	15,000 FALCON + 150 F-USDC
Borrow 10 F-USDC	□	Legacy `LoanSet` + broker co-sign
Repay 10.000137 F-USDC	□	Ledger **30899**, tx `71307F0B…`

7.8 E2E coverage matrix

Scenario	Coordinator script	Portal	Status
Vault supply	□ (optional step)	□	PASS
Permissionless borrow	□	□	PASS
Full installment repay	□	□	PASS
HF breach liquidation	□	Risk monitor UI	PASS
Payment-default liquidation	HF monitor	—	Not E2E’d (24h interval)
Broker legacy borrow + co-sign	—	□	PASS (2026-07-10)
`ClaimLPReward`	—	□	Prior sessions
Impair / unimpair only	HF monitor	—	Daemon logic; not isolated E2E

7.9 Live pool state (post-liquidation E2E, 2026-07-14)

Metric	Value
AMM FALCON	~47,984 FALCON
AMM F-USDC	~106.5 F-USDC
AMM price	~0.002219 F-USDC/FALCON (depressed by repeated E2E dumps)
Vault `AssetsTotal`	~257.6 F-USDC
Vault `AssetsAvailable`	~247.6 F-USDC

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8. Operations guide

8.1 Bootstrap (coordinator)

```
bash scripts/enable-lending-fleet.sh --wait
# After lending-permissionless docker image is built:
bash scripts/enable-lending-permissionless-fleet.sh --wait
python3 scripts/issue-testnet-stables.py
python3 scripts/bootstrap-testnet-lending.py
```

HF monitor:

```
bash scripts/deploy-lend-hf-monitor.sh
python3 /var/lib/qxrp-lending/lend-hf-monitor.py --once --dry-run
journalctl -u qxrp-lend-hf-monitor -f
```

E2E regression (coordinator):

```
python3 scripts/lend-e2e-permissionless.py
python3 scripts/lend-e2e-liquidation.py
```

8.2 Vercel / portal env

Variable	Required for	Notes
Testnet RPC	All	Overview + submit
`TESTNET_LENDING_BROKER_SECRET`	Legacy co-sign + HF daemon	Remove after permissionless live
`SIGNER_PROXY_URL` / `SIGNER_PROXY_TOKEN`	Legacy co-sign + daemon	
`LEND_HF_MONITOR_TOKEN`	HF monitor → `loan-manage`	Optional bearer auth

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9. Security model

Asset	Model
User Falcon seed	Passkey-encrypted in IndexedDB; browser-only signing
Broker owner secret	Server-only; legacy co-sign + daemon — not needed for permissionless borrow
Permissionless borrow	No gatekeeper; collateral + HF enforced on-ledger
Co-sign route	Testnet-only, origin-checked, `LoanSet` only (legacy)

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10. Known gaps

Remaining testnet / portal

- Payment-default liquidation E2E (wait for `PaymentInterval` + `GracePeriod` — impractical in quick scripts)
- Portal “Liquidate” button for third-party liquidators (today: HF monitor, `loan-manage` API, or coordinator scripts)
- Vercel redeploy; remove `TESTNET_LENDING_BROKER_SECRET` if legacy co-sign retired
- Live APY from epoch `EmissionRate` in overview
- AMM price recovery after repeated E2E dumps (pool depth testing)

Mainnet considerations

- Permissionless collateral-only borrow — no broker operator
- Do not bootstrap-mint F-USDC into vault (bridge-only)
- Liquidation under thin AMM depth / manipulation resistance
- HF monitor liveness and MEV on `LoanManage` default

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11. File reference

Portal (`qXRP-faucet-wallet`)

Path	Purpose
`src/app/lend/page.tsx`	Lend page orchestration
`src/components/lend/LendPanels.tsx`	UI panels
`src/app/api/lend/overview/route.ts`	Overview aggregation
`src/app/api/lend/borrow-preflight/route.ts`	Borrow preflight
`src/app/api/lend/loan-manage/route.ts`	LoanManage submit
`src/app/api/lend/cosign/route.ts`	Legacy broker co-sign
`src/lib/falcon-lend-tx-sign.ts`	Tx builders
`src/lib/lend-borrow-errors.ts`	Pre-flight + error copy
`src/lib/lend-collateral.ts`	HF math, min collateral
`src/lib/lend-loan-manage.ts`	LoanManage action helpers
`public/config/lending.json`	On-chain IDs + terms

Protocol (`qXRP`)

Path	Purpose
`src/libxrpl/tx/transactors/lending/LoanSet.cpp`	Borrow + permissionless path
`src/libxrpl/tx/transactors/lending/LoanManage.cpp`	Impair/default + `defaultPermissionlessLoan`
`src/libxrpl/ledger/helpers/LendingHelpers.cpp`	HF bps, collateral checks, AMM price
`scripts/enable-lending-permissionless-fleet.sh`	Amendment activation
`scripts/lend-hf-monitor.py`	HF enforcement daemon
`scripts/deploy-lend-hf-monitor.sh`	systemd install for HF monitor
`scripts/lend-e2e-permissionless.py`	Coordinator E2E: borrow + repay
`scripts/lend-e2e-liquidation.py`	Coordinator E2E: HF breach + default

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12. Conclusion

Falcon Ledger lending is a **verified vertical slice** on testnet:

- **`LendingPermissionless`** enabled fleet-wide; collateral-only `LoanSet` without broker co-sign
- **Coordinator E2E** confirms borrow, repay (6 dp ceil), and third-party liquidation via AMM price shock
- **HF monitor** running on coordinator for automated `LoanManage` enforcement

- **Portal** `/lend` wired for permissionless borrow, HF display, risk monitor, and preflight APIs

The most common user-facing confusion — **`tecINSUFFICIENT_PAYMENT` while holding ample F-USDC** — is a protocol requirement to pay **principal + interest/fees per installment**. The portal and E2E scripts surface the exact due amount (`5.000069` for a `5` F-USDC single-payment loan at 5% APR).

Next priorities: Vercel redeploy, optional portal liquidator UX, payment-default E2E, and mainnet hardening under realistic AMM depth.